

PANG, Zi Yang

+852 95799468 | zypang04@gmail.com | linkedin.com/in/zi-yang-pang/ | https://ziyang.is-a.dev

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Hong Kong, Sep 2023 – Dec 2027

BEng Computer Science + Extended Major in Artificial Intelligence + Minor in Business – First Class Honours (Expected)

- CGPA: 4.002/4.3 | ~ Top 1% of HKUST Undergraduates | 6x Dean's List | HKUST Full Scholarship Admission Holder

PROFESSIONAL EXPERIENCE

Zonic Tech – Full-Stack AI Engineer Intern

Hong Kong, June 2026 – Present

- Owned the entire architectural migration of the AI agent from *Vercel* and video-analysis from *Azure* into independent, containerized *Google Cloud Run* services orchestrated via *GCP Pub/Sub* and *Eventarc*, consolidating a fragmented multi-cloud architecture onto *Google Cloud Platform*, eliminating serverless timeouts and greatly reducing infrastructure cost.
- Built the company's video-editing AI agent on the *Claude Agent SDK*, engineering the skill library and custom tool suite that let it semantically search and retrieve analyzed footage information from *Upstash Vector* and *Redis*, then create edits autonomously.
- Scaled agent execution onto *Cloud Run Jobs*, extending max runtime by 45x to support our clients to process long-form footage.
- Secured interest from prospective clients and investors by pitching at an exhibition, receiving positive feedback and trial requests.

Zonic Tech – AI Engineer Intern

Hong Kong, June 2025 – Sep 2025

- Engineered an AI-powered video editing system that integrates OpenAI's *Large Language Model (LLM)* and *Retrieval-Augmented Generation* with *Upstash Vector* and *Redis* for dynamic data retrieval, leveraging *LangChain* to create an agentic AI workflow that helps users automate the video editing experience and deliver personalized results, enhancing user efficiency and engagement.
- Partnered with the CEO to deliver an engaging pitch to potential enterprise clients and investors, showcasing the company's groundbreaking video understanding model and its innovative applications in AI video editing, resulting in enthusiastic feedback.
- Built a custom AI state management system using nested *React* context architecture to replace Vercel's *AI SDK RSC* in the company's codebase, achieving approximately 60% faster client-side updates by shifting state handling to the client-side.

Comptify Analytics – Analyst Programmer Intern

Hong Kong, June 2024 – Aug 2024

- Spearheaded the development of an *SMTP*-based email verification tool using *Python* that identified 60% of invalid email addresses from the online-collected contact list, allowing the company to focus marketing efforts on the remaining 40% of potential clients.
- Engineered a full-stack director remuneration analytics platform that leverages market data to analyze compensation trends and deliver data-driven recommendations, empowering over 1,000 existing multinational enterprise clients to access executive pay insights and optimize their pay strategies, generating average fees of USD2,300 each, using *.NET*, *MySQL*, *jQuery*, and *Bootstrap*.

Zonic Tech – Frontend Developer Intern

Hong Kong, Dec 2023 – Jan 2024

- Revamped the user interface using *React.js*, enhancing user experience through an intuitive design and *Material UI* components.

EXTRA-CURRICULARS & LEADERSHIP EXPERIENCE

Morgan Stanley Code To Give Hackathon – Technical Lead

Hong Kong, Aug 2025

- Led a 6-developer team to build a digitised learning hub for children, parents, and teachers using *React*, *Flask*, and *PostgreSQL*.
- Engineered a real-time chat system using *React* and *Supabase* to facilitate instant advice-sharing for a community of parents.

Hong Kong Malaysian Student Association – Lead Webmaster

Hong Kong, Jan 2025 – Dec 2025

- Led a 5-developer team to design and deploy an intuitive events page to showcase the organization's annual events, utilizing *Next.js*, significantly streamlining information access and enhancing user understanding of organizational initiatives.
- Reviewed and debugged *Next.js* code authored by webmasters, managing *Git* workflows for clean merges and stable deployments.

HKUST Computer Science Department – Undergraduate Teaching Assistant

Hong Kong, Jan 2025 – Present

- Mentored 510 students in C++, improving their grasp of syntax, programming logic, and core concepts like pointers and OOP.

PROJECT EXPERIENCE

MatesMarket (HKUST online marketplace) [www.matesmarket.online] – Co-Founder & Full-Stack Developer

- Engineered a HKUST-exclusive second-hand trading platform using *Next.js*, *TypeScript*, *Firebase*, and *TanStack Query*, growing to over 150 users within one month of launch.
- Pitched and presented the application to university officials, securing their support and endorsement for the initiative.

SKILLS

● Technical:

- Languages & Frameworks – Python, C++, SQL, JavaScript, HTML/CSS, Flask, PostgreSQL, SQLite, Next.js, React.js
- Developer Tools & Platforms – Firebase, Supabase, Google Cloud Platform, Microsoft Azure, Postman, Git

- **Languages:** English (Native), Mandarin (Fluent), Cantonese (Fluent), Malay (Fluent), Hakka (Basic)